

Prediction of Lung Cancer Risk through Machine Learning based on Lifestyle Questionnaire Data

1st Samir Diego Chávez Cáceres 2nd Angel Eduardo Hinocho Jove 3rd Maribel Molina 4th Aurea Soriano-Vargas
Universidad Nacional Universidad Nacional Universidad Nacional Universidade Estadual
de San Agustín (UNSA) de San Agustín (UNSA) de San Agustín (UNSA) de Campinas (UNICAMP)
schavezca@unsa.edu.pe ahincho@unsa.edu.pe mmolinab@unsa.edu.pe asoriano@unicamp.br

Abstract—Lung cancer is the most common cause of mortality among various types of cancer globally, hence the pursuit of alternative methods for early identification of lung cancer should be a significant consideration, especially in low-resource settings. In this work, we present an alternative for lung cancer risk recognition based on machine learning. Utilizing lifestyle questionnaire data, we collected, analyzed, visualized, and processed data to train machine learning models. After balancing the data through SMOTE, our proposed model, XGBoost, was fine-tuned using the GridSearchCV method, to which we allocated 70% of our data, leaving the remainder for testing. The results obtained by the model for the first dataset included accuracy and F1 score of 96.50%, with precision and sensitivity of 96.51%. For the second dataset, an accuracy of 95.83%, F1 score of 95.83%, precision of 96.27%, and sensitivity of 95.83% were achieved. Additionally, through LIME, we obtained local interpretability of predictions, thus contributing to a more transparent understanding of model behavior and providing a clearer perspective on machine learning-based solutions.

Index Terms—Lung Cancer, Machine Learning, Lifestyle, Explainability, Exploratory Data Analysis, Classification.

I. INTRODUCTION

Lung cancer stands as the primary cause of mortality among various cancer types globally. Based on 2020 statistics, the incidence of deaths attributable to this pathological entity reached 1.8 million, constituting a concerning 22% of total fatalities associated with malignant neoplasms [14].

Thoracic computed tomography (CT) and chest radiography are conventional methods for lung cancer diagnosis. In specific situations, magnetic resonance imaging (MRI) and positron emission tomography (PET) may be employed for staging to assess the extent of cancer spread, thereby contributing to determining the most appropriate therapeutic strategy. Bronchoscopy and biopsy, either through needle aspiration or surgical procedures, are imperative for achieving accurate lung cancer diagnosis and providing information regarding histological type [13].

However, it is imperative to consider their application, especially in the context of early-stage disease. These methods are particularly beneficial in advanced stages of lung cancer, where identifying the extent of disease spread is crucial for determining the most suitable therapeutic approach [10]. Nevertheless, their utility in the early stages of the disease may be limited due to the possible absence of specific symptoms

justifying their application, potentially leading to unnecessary use of these procedures.

Early identification of lung cancer plays a crucial role in improving survival rates. However, the absence of symptomatic manifestations in the early stages of this disease poses significant challenges in its early detection [15].

Moreover, repeated exposure to radiation, especially in medical imaging procedures, poses potential health risks such as genetic mutation induction and increased risk of secondary cancer [5]. Therefore, the benefits of these methods must be carefully weighed against potential risks, especially in individuals who may require periodic evaluations.

Additionally, the economic and geographical accessibility of these procedures presents a significant barrier, especially in low-resource communities. These challenges underscore the need to consider more strategic approaches in lung cancer diagnosis, especially in resource-limited environments.

Currently, machine learning (ML) techniques play a crucial role in the healthcare domain. For lung cancer specifically, numerous scientific studies have been conducted from the perspective of machine learning. The results of studies conducted to date indicate that machine learning has the potential to improve the early detection of lung cancer [4], [8], [9].

However, limitations are present in many previous works. A notable observation reveals that a significant portion of existing studies focuses on black-box algorithms, posing significant challenges in terms of explainability and understanding. It is common to find insufficiently defined model training protocols, raising concerns about the reproducibility and generalizability of results. Lack of transparency in training methodology can affect the validity of conclusions, as details regarding data selection, hyperparameter tuning, and model validation are essential to ensure the robustness and applicability of implementation in clinical settings.

Furthermore, there is a trend towards omitting analysis of intrinsic dataset characteristics in many previous works. Limited evaluation of data quality, imbalanced class distributions, or lack of attention to potential biases may compromise the validity and applicability of developed models. Interpreting results without a comprehensive analysis of these underlying characteristics may lead to misleading conclusions.

In this work, we propose a methodology for lung cancer risk prediction using machine learning classification models based

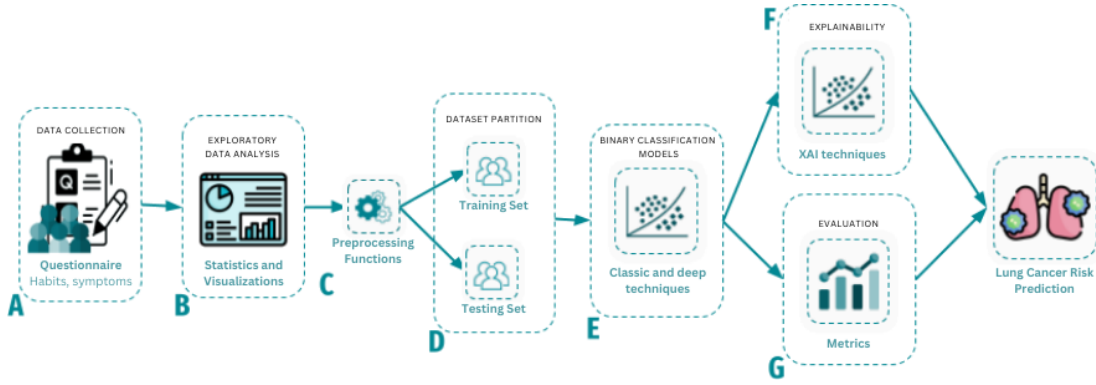


Fig. 1: General Description of the Methodology: The questionnaire data (A) undergo processing and formatting into a tabular structure to facilitate utilization by subsequent algorithms. Subsequently, an exploratory data analysis (B) is conducted on the tabulated data, serving as the basis for problem identification and selecting requisite preprocessing functions (C). The processed data is then partitioned into training and testing sets using the Hold-Out method (D), followed by training of binary classification models (E) training. These trained models are subsequently employed alongside explainability techniques (F) and evaluated using pertinent metrics (G) to ultimately predict the risk of lung cancer, substantiated by adequate justification.

on lifestyle questionnaire data including common habits and symptoms. Our contribution lies in conducting a comparative evaluation among various classifiers to develop a classification model exhibiting maximum sensitivity and discriminative capability for individuals at high risk. This strategy aligns with the aspiration to achieve early and accurate detection, thereby optimizing medical interventions and improving treatment prospects. Additionally, our methodology not only considers the predictive efficacy of models but also places particular emphasis on result explainability. Furthermore, we analyze and characterize the data, addressing aspects such as information quality, class distribution, and potential biases.

II. RELATED WORK

To enhance the early detection of lung cancer, numerous studies have focused on machine learning models employing data collected from specific questionnaires. This strategy offers a comprehensive insight by integrating information on habits, medical history, and risk factors, key elements for accurate and early detection of this disease.

Some authors have concentrated on developing predictive models for lung cancer prediction using datasets available on Kaggle, exploring various classification ensemble techniques, including XGBoost and LightGBM. Mamun et al. [7] worked by comparing four classification models: XGBoost, LightGBM, Bagging, and AdaBoost, utilizing 10-fold cross-validation. They utilized an imbalanced dataset available on Kaggle and opted to use the SMOTE oversampling technique. According to their results, the XGBoost technique outperformed the other three models, achieving an accuracy of 94.42%. Rikta et al. [12] used three classification models: XGBoost, LightGBM, and GBM. They also utilized a dataset published on Kaggle and applied Random OverSampling (ROS) as a balancing technique. The authors took their analysis a step further by employing Principal Component Analysis (PCA) for selecting relevant features and used GridSearchCV to obtain

optimal model hyperparameters. To explain the results, the authors employed the SHAP tool. Reported results indicate that the GBM classification technique outperformed the other two models, achieving an accuracy of 98.76%.

On the other hand, significant efforts have been made to rely on classical machine learning strategies to predict lung cancer probability. Dritsas et al. [3] employed various classification models, including Naive Bayes (NB and BayesNet), Logistic Regression (LR), Logistic Model Tree (LMT), SVM, SGD, J48, Random Tree (RT), Rotation Forest (RotF), Reduced Error Pruning Tree (RepTree), RF, AdaBoostM1, multi-layer perceptron (MLP), and K-NN. They evaluated accuracy, precision, sensitivity, F-Measure, and AUC metrics. The authors applied the SMOTE technique and conducted a 10-fold cross-validation on a dataset available on Kaggle. Reported author results indicate that the RotF technique demonstrated the best performance among all models, with accuracy 97.1%. Patra R. [8] analyzed various classification models using a dataset available on the “UC Irvine Machine Learning Repository”. The author evaluated four classification models: KNN, Naive Bayes, RBF, and J48. Using the Weka tool, a 10-fold cross-validation was conducted. The results indicate that the RBF algorithm outperformed all other algorithms with an accuracy of 81.3%. Likewise, Ahmad and Mayya [1] used Weka to build decision trees and random forest models to process data from a public dataset on Kaggle. The obtained results were 93.33% of accuracy. Also using a machine learning tool, Viera et al. [17] proposed an Artificial Neural Network model from a group of models trained through the CRISP-DM methodology and RapidMiner software. This model group underwent a series of scenarios where the best model achieved an accuracy of 93%. The authors used a public dataset on Kaggle.

Additionally, significant efforts have been made to leverage deep learning strategies as a frontier in research. Thaventhiran & Sekar [16] developed a new technique called TPFMDANN-

LSTM (Target Projection Feature Matched with Deep Neural Networks with LSTM). This Deep Learning model consists of an input layer receiving data and another layer showing results, with two hidden layers in between; one for selecting the best features and the other for proper classification using LSTM. The authors tested their model using a dataset from Kaggle and DataWorld, comparing the results with two other methods (LCP-CNN and K-NN). Their proposed model demonstrated better performance in terms of accuracy, fewer false positives, and reduced time consumption.

Overall, these works demonstrate that machine learning has the potential to improve early detection of lung cancer, have a solid methodology, and do not omit dataset characteristics. However, most works ignore explainability issues [1], [7], [8], [16]. Omitting explainability in machine learning not only hinders model transparency but also obstructs understanding and trust in model predictions. This can result in a poor user experience, as they face a process perceived as “magic”. Unfortunately, only a few works provide an adequate explanation of the datasets used, either during the analysis phase as observed in [2], [3], [17] or by including an explainability report for each prediction generated by the model as seen in [12].

III. METHODOLOGY

The methodology begins with the preparation of data from the questionnaires (Figure 1.A), which are processed and transformed into a tabular format to facilitate their use by subsequent algorithms. Subsequently, exploratory data analysis is conducted (Figure 1.B), from which potential issues are identified and necessary preprocessing functions (Figure 1.C) are selected to improve the quality and relevance of the data.

The processed data is split into training and testing sets using a cross-validation strategy (Figure 1.D), allowing for robust evaluation of model performance. Next, binary classification models are trained (Figure 1.E).

Subsequently, explainability techniques are incorporated (Figure 1.F) to provide clarity on how the models make decisions. This not only contributes to understanding the results but also facilitates the identification of potential biases or areas for improvement in the model.

Model evaluation is conducted using specific metrics (Figure 1.G), providing a quantitative measure of performance in terms of accuracy, sensitivity, and F1 score, among others. This metric analysis contributes to the objectivity and validity of the results obtained.

Finally, based on the trained models, the incorporated explainability, and the evaluative metrics, the prediction of lung cancer risk is made. This approach ensures the generation of predictions supported by solid arguments and transparent justifications, thus substantiating the clinical utility of the conclusions drawn.

A. Data Collection

In this study, two datasets from Kaggle were utilized, both compiled from questionnaires and medical annotations. The

first dataset¹ has 309 instances, 15 variables, and a binary dichotomous label for lung cancer. Some of these attributes describe patient habits such as smoking or alcohol consumption. It also includes attributes related to mental state such as anxiety and peer pressure.

Regarding the second dataset², it consists of 1000 instances, 23 attributes, 2 identifiers (index and patient id), and a label related to the level of cancer risk. Some attributes describe habits such as alcohol consumption, obesity, smoking, or passive smoking, as well as attributes related to the environment such as air pollution or occupational risk.

B. Exploratory Data Analysis and Data Preprocessing

We focus on addressing the question: i) How do habits influence the risk of developing lung cancer? For this purpose, we identified certain generalities of both datasets and finally performed a comparison between them.

1) *Dataset # 1:* All attributes are binary categorical except for age; there are no null or univalent values, and there are 33 duplicate values. Additionally, the dataset is highly imbalanced. The age attribute follows a log-normal distribution (see Figure 3.a), which facilitates how we handle our data. The class attribute “lung cancer” is a binary categorical variable indicating whether there is a risk of lung cancer or not.

Regarding the categorical attributes, they can be divided into three types: habits, mental state, and symptoms. Among the habit attributes, such as alcohol consumption and smoking, it is noteworthy that all female individuals who consume alcohol are at risk of developing lung cancer, and the majority of male individuals who consume alcohol are also at risk of developing lung cancer. This can be observed in Figure 2.a. Also women who present the mental state attributes appear to be at risk of developing lung cancer Figure 2.a.- 2.b.

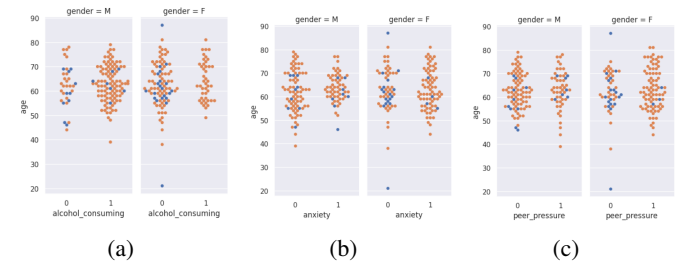


Fig. 2: Swarm plots of the risk of developing lung cancer: (a) based on alcohol consumption; (b) based on age and anxiety, by gender; and (c) based on age and peer pressure, by gender. Lung cancer people in orange and healthy people in blue.

2) *Dataset # 2:* All attributes are categorical (scales from 1-9) except for age. There are no null or univalent values, and the presence of 848 duplicate values can be highlighted. Despite this, the dataset is slightly imbalanced. Regarding age, it is observed that it fits a beta distribution, as shown in Figure

¹<https://www.kaggle.com/datasets/mysarahmadbhat/lung-cancer/data>

²<https://www.kaggle.com/datasets/thedevastator/cancer-patients-and-air-pollution-a-new-link/data>

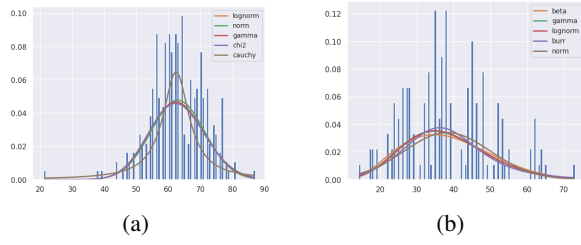


Fig. 3: Distribution of the age attribute within (a) the first dataset; and (b) the second dataset.

3.b. The class attribute "level" label is a ternary categorical variable indicating the level of lung cancer risk (low, medium, and high).

The categorical attributes can be divided into three types: habit, environmental, and symptom attributes. Among the habits, such as alcohol consumption, smoking, passive smoking, and obesity, it is noteworthy that the majority of individuals with a medium or high risk of developing lung cancer are those who consume alcoholic beverages to a greater extent, as depicted in Figure 4.

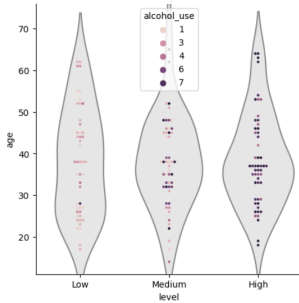


Fig. 4: Violin plot of the risk of developing lung cancer concerning alcohol consumption and age.

Additionally, compared to the first dataset, we have two more habits: obesity and passive smoking. Regarding obesity, individuals with a higher risk of developing lung cancer generally exhibit a medium or high level of obesity. Similarly, the habit of passive smoking poses a high risk of developing lung cancer. Both cases can be observed in Figures 5.a and 5.b, respectively.

3) *Comparative Analysis*: In both datasets, we can observe the influence of alcohol consumption, smoking habits, and obesity in individuals at high risk of developing lung cancer. We can also find influence from other factors, such as mental states which also show a relationship with this risk.

All these observations will aid in preprocessing the data in both datasets. To deal with the imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) will be used, where the necessary synthetic samples from the minority class will be generated.

C. Dataset Partitioning

Once the dataset is balanced, we can select a partitioning method. The method used is Hold-Out with 70% of the data

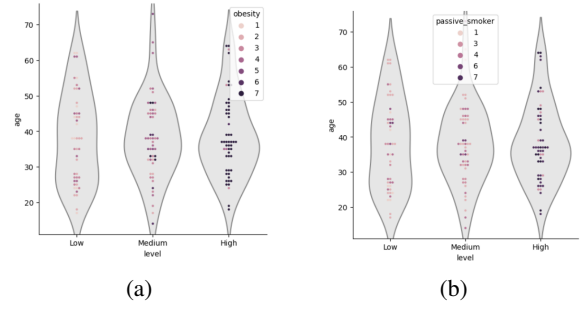


Fig. 5: Violin plots of the risk of developing lung cancer concerning: (a) obesity and age; and (b) passive smoking and age.

for training and 30% for testing. Additionally, for the training data, we will apply 5-fold cross-validation for model fitting. The k -fold cross-validation method is a technique for small-sized datasets, where the main idea is to divide the dataset into k blocks of the same size [6].

We then use one of these blocks for validation and the remaining blocks for training. This process is repeated k times, with the validation block changing to the next one in each iteration until iterating through all blocks. After this process, the result average for each metric to be reported is taken.

D. Classification Models

Once the partitioning phase is completed, we can train a machine-learning model. Below, we explain the different models chosen for the classification task.

A total of 7 classification models were applied, including classical and deep techniques. The GridSearchCV technique was used to obtain the best hyperparameters. This technique consists of a grid with all the parameters to be tested within a model.

Then, an algorithm is executed to choose a set of hyperparameters that will be used to train a model. Finally, through the k -fold cross-validation method, estimated metrics are obtained, which will be used to compare with other sets of hyperparameters and thus obtain the best set.

The selected classification models were: Support Vector Classifier (SVC), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), XGBoost (XGB), Stochastic Gradient Descent (SGD), and Artificial Neural Networks (ANN).

E. Explainability and Evaluation

Once the models are trained using the datasets, we proceed to evaluate their generalization power using metrics. These metrics will be useful for comparing the generalization performance of each model.

Additionally, using the LIME method, we provide a brief explanatory description of the prediction result. LIME (Local Interpretable Model-agnostic Explanations) [11] is a technique that explains predictions of various classifiers in an interpretable manner by learning a local model around the prediction.

TABLE I: Metrics reported by each model trained on the two datasets.

Results of Dataset I				
Model	Accuracy	F1-score Weighted	Precision Weighted	Recall Weighted
SVC	0.9580	0.9580	0.9580	0.9580
LR	0.9650	0.9650	0.9651	0.9650
DT	0.9091	0.9090	0.9132	0.9091
RF	0.9720	0.9720	0.9720	0.9720
XGB	0.9650	0.9650	0.9651	0.9650
SGD	0.9580	0.9580	0.9584	0.9580
ANN	0.9441	0.9440	0.9444	0.9441

Results of Dataset II				
Model	Accuracy	F1-score Weighted	Precision Weighted	Recall Weighted
SVC	0.9583	0.9583	0.9627	0.9583
LR	0.9583	0.9583	0.9627	0.9583
DT	0.9167	0.9167	0.9167	0.9167
RF	0.9583	0.9581	0.9625	0.9583
XGB	0.9583	0.9583	0.9627	0.9583
SGD	0.9167	0.9158	0.9325	0.9167
ANN	0.9167	0.8546	0.8562	0.8542

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The performance of the models (SVC, LR, DT, RF, XGB, SGD, ANN) within each lung cancer dataset has been computed in Table I for both datasets. For the comparison between each model, we present the results according to accuracy, F1 score, precision, and recall. The metrics were calculated considering the size of each class (weighted).

We can observe that in the case of the first dataset, the Random Forest (RF) model achieved the highest accuracy of 97.20% and F1 score (weighted) of 97.20%. Following this outcome, XGB and LR demonstrated strong performance. For predicting the risk of cancer within the first dataset, RF performed satisfactorily in terms of accuracy and F1 score.

In the case of the second dataset, the XGBoost (XGB), Logistic Regression (LR), and SVC models achieved an accuracy of 95.83% and an F1 score (weighted) of 95.83%. These models also demonstrated satisfactory performance in predicting the risk of cancer within the second dataset.

Additionally, through the LIME tool, it is possible to locally explain the prediction results of the different models.

For the first dataset, in Figure 6, we have the explainability of a prediction from the RF model, with the 5 most influential attributes being: allergy, alcohol consumption, difficulty eating, chronic disease, and coughing. From the feature/value table, we can find that the gender of the person is Female (0), they are 74 years old, and they have the condition of peer pressure.

Additionally, the features that had the most influence on the XGB and LR model prediction were: allergy, difficulty eating, fatigue, alcohol consumption, coughing, chronic disease, and peer pressure.

From these explanations, we can find certain coincidences. Alcohol consumption and coughing are identified as factors influencing the predictions (Figure 6). This aligns with [3], [12], which consider coughing and alcohol consumption among the

top 7 characteristics that most influenced the local explainability of their predictions. Additionally, we find anxiety states are the most influential characteristics.

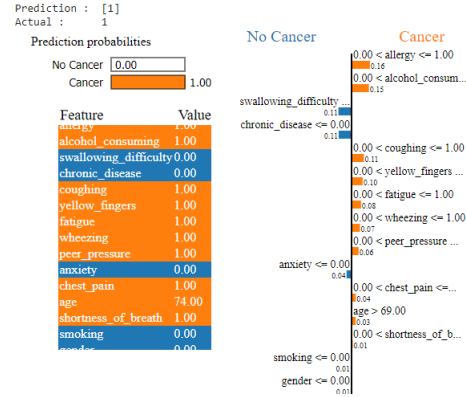


Fig. 6: Lime Explainability for RF model from the first dataset.

In the case of the second dataset: In Figure 7, we observe that the 5 most influential attributes in the XGB prediction were: difficulty breathing, clubbing (abnormal nail bed shape), alcohol consumption, passive smoking, and smoking. Alcohol consumption and clubbing are the most valuable attributes (8 and 9) in predicting a low risk of lung cancer.

The features that had the most influence on the LR and RF model prediction were: alcohol consumption, difficulty breathing, obesity, fatigue, coughing blood, snoring, and a balanced diet.

From these explanations, we can highlight that alcohol consumption and difficulty breathing are the factors that influence the predictions, which aligns with [3], [12], who consider both attributes among the top 7 influential characteristics in their model decisions, and [1] who consider them as medium-risk factors for lung cancer.

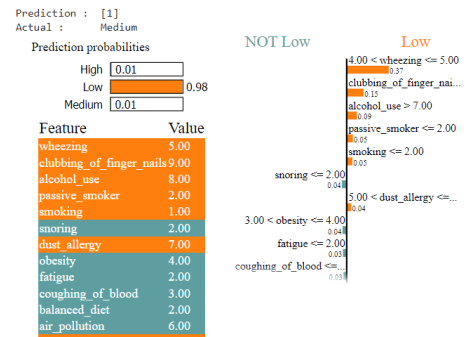


Fig. 7: Lime Explainability for XGB model from the second dataset.

Regarding the performance of each model, our XGB, RF, and LR models, which achieved an accuracy of 96.50% / 95.83%, 97.20% / 95.83%, and 96.50% / 95.83%, respectively, outperformed their counterparts in the related work [1], [3], [7], [12] except for the model by Ding et al. (2023) [2].

On the other hand, ANN achieved an accuracy of 94.41% and 91.67%, which performed similarly or better, in the case

of the first dataset, to the model presented by Dritsas et al. (2022) [3] and Viera et al. (2021) [17]. Regarding the second dataset model, its performance is surpassed by the model of Thaventhiran et al. (2022) [16].

The predictions of each model were benefited by the local explainability offered by LIME. It is interesting to note how habits such as alcohol consumption, passive smoking, and obesity significantly influence the predictions (Figure 6). Also, LR, XGB, and RF are among the best models, coinciding within the literature [1]–[3], [7].

Finally, considering the coincidences with the mentioned authors, we decided to propose the XGBoost (XGB) model, due to its overall results in both datasets. As a second alternative, the Random Forest (RF) and Logistic Regression (LR) models can also be considered. It is important to mention that the choice of a machine learning model must always be aligned with the data analysis (EDA), our proposed model should be considered as an alternative and not as the definitive solution.

V. CONCLUSIONS AND FUTURE WORK

Lung cancer is the most common cause of mortality among various types of cancer globally. Factors such as radiation exposure, and economic and/or geographical accessibility to traditional procedures pose a significant barrier to the early identification of lung cancer, especially in low-resource settings.

In this study, we present an alternative methodology for recognizing the risk of lung cancer, which is based on machine learning along with the data science process. The data were collected from the public repository Kaggle, where we dealt with certain peculiarities such as imbalance and duplicate data. Subsequently, relationships among the data were found, which were discovered through the process of exploratory data analysis. Thus, the decision was made to divide the data into 70% for model training and 30% for testing. Finally, in the training data, the 5-iteration cross-validation technique along with GridSearch was used to find the best hyperparameters for the various models to be trained.

The models chosen for training included: Support Vector Classifier, Logistic Regression, Decision Tree, Random Forest, XGBoost, Stochastic Gradient Descent, and Artificial Neural Networks. Based on the results and current literature, the XGBoost model is proposed, which achieved an accuracy, F1 score of 96.50%, precision, and sensitivity of 96.51% in the first dataset. In the second dataset, it achieved an accuracy of 95.83%, F1 score of 95.83%, precision of 96.27%, and sensitivity of 95.83%.

Additionally, through the LIME tool, we can obtain local explainability for each prediction made by our machine-learning model. This allows us to observe the primary influence of unhealthy habits such as alcohol consumption, smoking, and obesity on the models' predictions. This enables us to have a clearer understanding of the models' behavior and provide a clearer perspective on solutions based on machine learning.

Our proposed method aligns to achieve early and accurate detection, thereby optimizing medical interventions and improving treatment prospects with a particular emphasis on the explainability of the results obtained. Despite the promising results of our model, we hope to achieve better results with a larger dataset. In the future, we aim to develop a web or mobile application to deploy our proposed model.

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